

A008 · A fast and memory-efficient framework for similarity networks in biology

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Similarity networks have seen widespread application in biology. A similarity network is a graph where nodes represent biological entities and edges are drawn between nodes with similarity scores above a given threshold. Examples include Blast scores between protein sequences, TM-scores between protein structures, and neighborhood similarity scores between gene loci. Similarity networks are useful for grouping the vastness of biological diversity into clusters (connected components in the graph) that can be individually investigated and compared to each other. Similarity networks can comprise tens or hundreds of thousands of nodes, presenting computational challenges for efficient generation, storage, and visualization. In this work, we suggest storing networks as maximum spanning trees (MSTs) in sparse matrices, with the optional addition of k-nearest neighbor edges (KNN). We develop software for generating similarity matrices for a variety of input file types and distance metrics, and for visualizing and exploring the resulting networks.